EthnoHack 2026

EthnoLens AI

Historical plant knowledge, kept beside the passage it came from.

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github.com/Jeta-Syla/EthnoLens-AI

Not medical advice: described uses are historical records.

Plant knowledge is locked in old pages

- Plant knowledge sits in manuscripts, old books and field studies. Names are historical and the spelling changes: a medieval recipe says “sauge”, not sage.
- A researcher reads page after page, decides whether a word is a plant, and keeps the link to the folio by hand.
- The source is easy to lose once a page is summarised. Local names, including Albanian names from Kosovo and Albania, are at risk of being forgotten.



Four steps, each one checkable

Scan & Discover

Upload a page. The plant name comes back marked inside its passage.

Archive

Search by name, part, preparation, source and language. What the text states is kept apart from what is only a candidate.

EthnoMap

A point only for a documented region in Kosovo or Albania, with the Albanian name and its source.

Ask

Answers only with a passage you can open. No source: “no data”. Dose or treatment questions are refused.

Every record keeps its **source, folio, passage, extraction confidence and mapping confidence**, and whether the name is **stated** in the text or **inferred**.

Example

From one manuscript line to a record

MS Add. 9308, f. 5v (c. 1400): *“For þe coughe . tak **sauge** and rue and comyn and pouder of peper and seth hem to gedere in hony and mak a letuarie …”*

Historical name (as written)	sauge — stated in the text, its position marked
Normalised · scientific name	sage · Salvia officinalis L. — inferred, checked with GBIF
Albanian local name	Sherbela (Mustafa et al. 2012, Albanian Alps, Kosovo)
Preparation · described use	“seth hem to gedere in hony and mak a letuarie” · “For þe coughe”
Plant part	empty — the passage does not say, so nothing is guessed
Source	MS Add. 9308, f. 5v · review status: pending

What you will see

Scan: upload a page; sauge, rue, comyn and peper are marked in the passage, and sauge is linked to Sherbela.

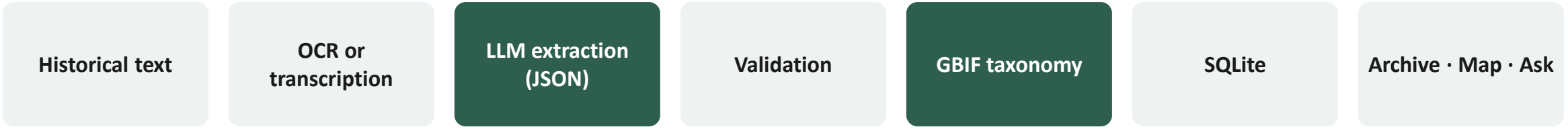
Archive: 200 records from 100 recipes; open “sauge” as a specimen sheet: candidate scientific name, Albanian name, confidence.

EthnoMap: a point only for a documented region; the popup shows the Albanian name and the source.

Ask: “Which sources mention sage?” returns cited passages that say “sauge”. A dosage question is refused; a question with no supporting passage gets “no data”.

Every page carries the warning: described uses are historical records, not instructions.

From manuscript to verifiable record



Interface: Streamlit; the map is Leaflet through folium.

API: FastAPI — /records, /documents, /plants, /map, /ask.

Extraction: LLM on Groq, temperature 0; each record validated with Pydantic and its position checked in the passage.

Taxonomy: GBIF species match; only an exact species match is “high”, near-misses stay “candidate”. Albanian names (Mustafa et al. 2012) are tied to the accepted name.

Reproducible: one command rebuilds the database from the original sources.

What we measured

80.9%
extraction precision (50 annotated recipes)

60.8%
recall · F1 69.4%

100%
of 200 records traceable to their passage

2.7%
OCR character error rate on 25 BHL pages

15/15
medical questions refused (English and Albanian)

100
Albanian names for 98 species, linked through GBIF

Data: 100 recipes from MS Add. 9308 (Middle English, c. 1400, Curious Cures, Cambridge). Test set: 50 recipes annotated by the team. Ask citation coverage: 100% (3/3 historical questions). Manual review: 10/10 passed. Chroma: 100/100 passages indexed. Corpus: 183 transcribed pages, 100 recipes, 25 BHL OCR pages.

Honest by design

1. **No medical advice:** No dose or treatment. The warning stays on every page, and Ask refuses the question.
2. **Candidates are not facts:** A candidate scientific name is never shown as a confirmed identification.
3. **Stated vs. inferred:** What the text says and what the model infers are kept apart.
4. **Credited sources:** Local names come only from published sources, with a citation.
5. **Licences respected:** Texts with an unclear licence are not redistributed: the pipeline downloads them from the source.

Who it is for, and why it matters

Researchers

Ethnobotany, history of medicine, digital humanities: every source that mentions a plant, in minutes.

Students

Historical sources next to modern botany, with what the text says kept apart from what is inferred.

Libraries & museums

Collections become searchable and connected, without copying restricted texts.

The public

Local plant heritage, including Albanian names from Kosovo and Albania, kept with its source.

Knowledge preserved in documents of the past does not stay locked in archives: it becomes structured, connected and verifiable.

What comes next

1

More Balkan sources

Kosovo, Albania and the Balkans, and a map that grows only with documented regions.

2

Handwriting

Handwritten text recognition for pages a user uploads.

3

Review tools

For researchers, students and heritage institutions to verify or reject records.

4

Larger evaluation

More annotated recipes, with the hand-annotated set reported separately.

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